

1.Covariance and correlation

CODE:

```
# Photograph preference data
```

```
A <- c(18, 2, 20)
```

```
B <- c(22, 28, 10)
```

```
C <- c(20, 40, 40)
```

```
# Create data frame
```

```
photo <- data.frame(A, B, C)
```

```
# 1. Sample covariance between B and C
```

```
cov(B, C)
```

```
# 2. Sample covariance matrix
```

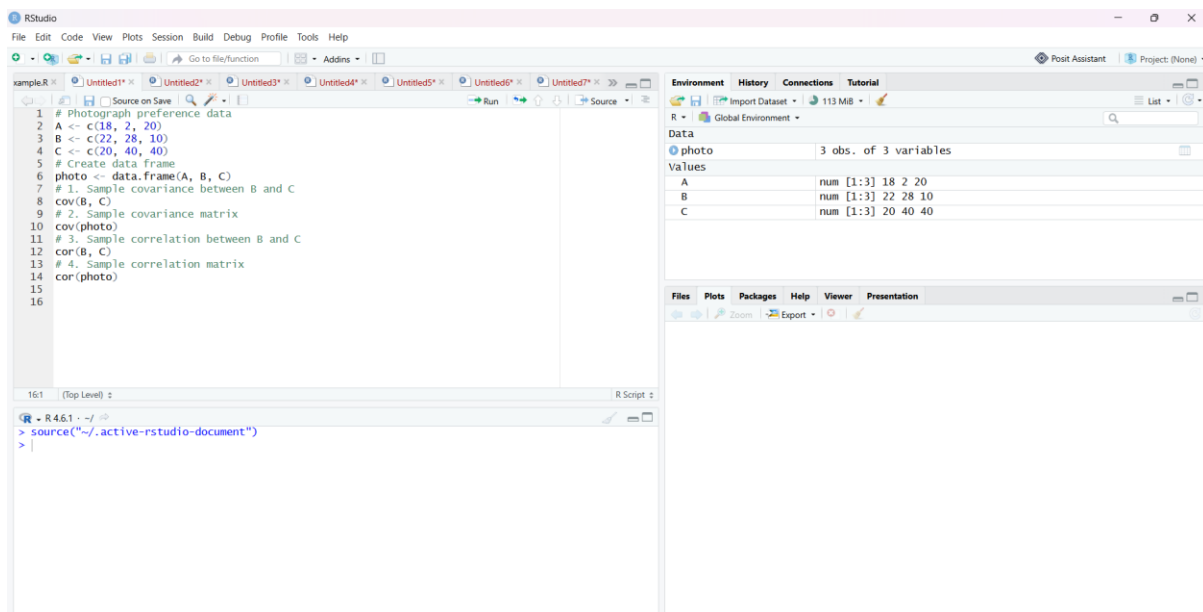
```
cov(photo)
```

```
# 3. Sample correlation between B and C
```

```
cor(B, C)
```

```
# 4. Sample correlation matrix
```

```
cor(photo)
```

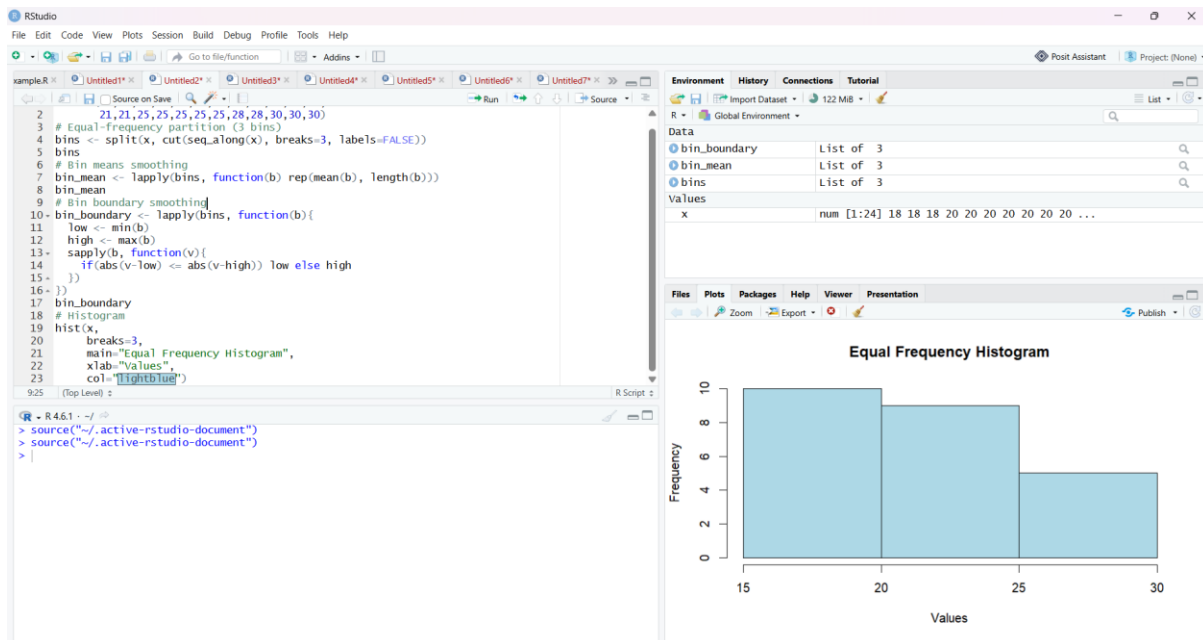


2. All Electronics data warehouse for analysis, Plot Histogram

```
x <- c(18,18,18,20,20,20,20,20,20,20,21,21,  
      21,21,25,25,25,25,25,28,28,30,30,30)
```

CODE:

```
# Equal-frequency partition (3 bins)  
bins <- split(x, cut(seq_along(x), breaks=3, labels=FALSE))  
bins  
  
# Bin means smoothing  
bin_mean <- lapply(bins, function(b) rep(mean(b), length(b)))  
bin_mean  
  
# Bin boundary smoothing  
bin_boundary <- lapply(bins, function(b){  
  low <- min(b)  
  high <- max(b)  
  sapply(b, function(v){  
    if(abs(v-low) <= abs(v-high)) low else high  
  })  
})  
bin_boundary  
  
# Histogram  
hist(x,  
      breaks=3,  
      main="Equal Frequency Histogram",  
      xlab="Values",  
      col="lightblue")
```



3. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows:

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 84

Class B: 51, 56, 84, 60, 59, 70, 63, 66, 50

(i) Find which class had scored higher mean, median and range.

(ii) Plot above in boxplot and give the inferences

CODE:

```
A <- c(76,35,47,64,95,66,89,36,84)
```

```
B <- c(51,56,84,60,59,70,63,66,50)
```

Mean

```
mean(A)
```

```
mean(B)
```

Median

```
median(A)
```

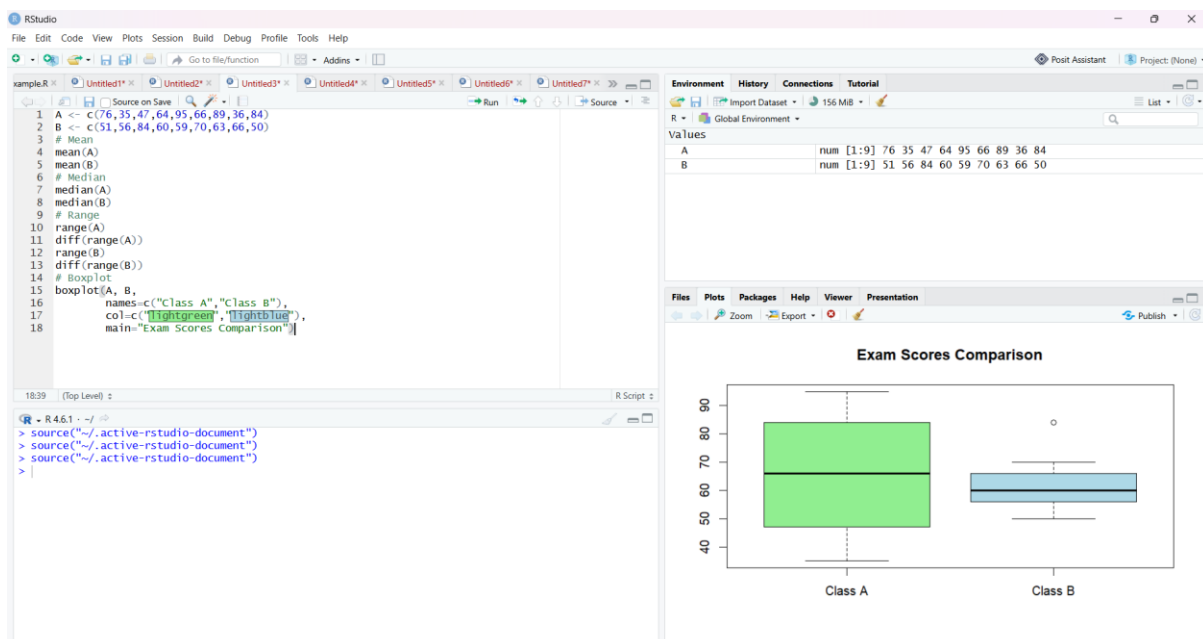
```
median(B)
```

Range

```

range(A)
diff(range(A))
range(B)
diff(range(B))
# Boxplot
boxplot(A, B,
        names=c("Class A", "Class B"),
        col=c("lightgreen", "lightblue"),
        main="Exam Scores Comparison")

```



4.Normalization

CODE:

(a) Min-max normalization example

```
minF <- 50000
```

```
maxF <- 100000
```

```
v <- 80000
```

```
normalized_v <- (v - minF) / (maxF - minF)
```

```
print(normalized_v)
```

(b) Normalize the data

```
x <- c(200, 300, 400, 600, 1000)
```

Min-max normalization (0 to 1)

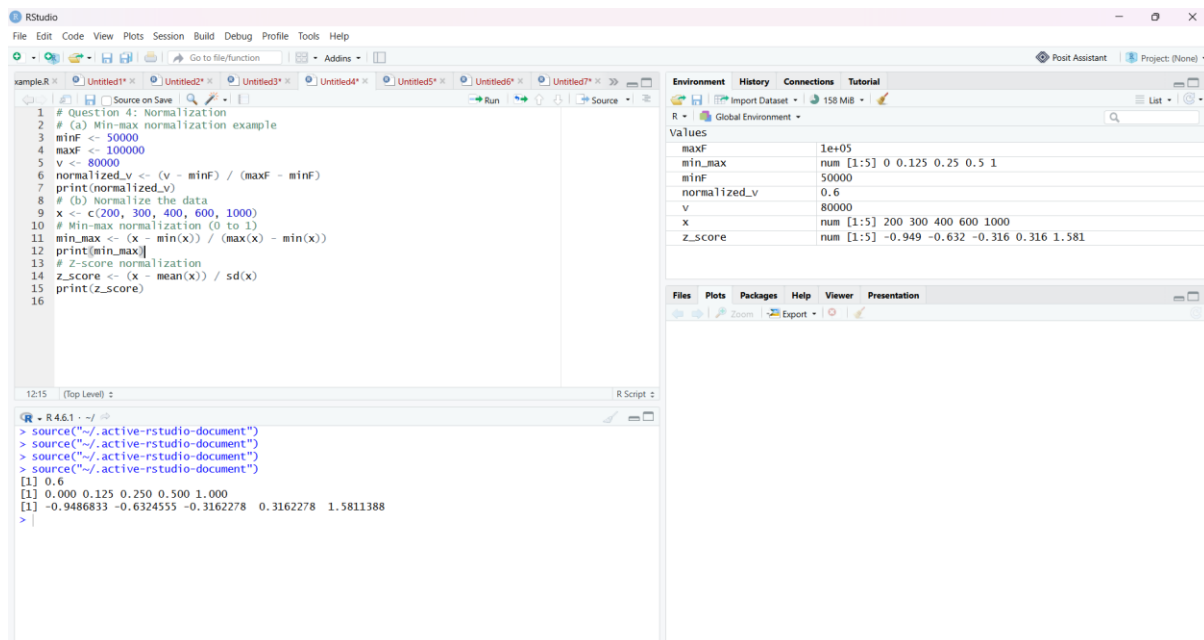
```
min_max <- (x - min(x)) / (max(x) - min(x))
```

```
print(min_max)
```

Z-score normalization

```
z_score <- (x - mean(x)) / sd(x)
```

```
print(z_score)
```



5. Histogram for the “AirPassengers

CODE:

```
data(AirPassengers)

hist(AirPassengers,
     breaks = seq(100, 700, by = 150),
     xlim = c(100, 700),
     col = "lightblue",
     main = "Histogram of AirPassengers",
     xlab = "Number of Passengers",
     ylab = "Frequency")
```

